

Top 10 Packet Tracer Lab Breakers

Professional Troubleshooting Edition

1. Wrong Cable Types

Problem: Links show red/down or devices fail to communicate.

Possible Reasons:

- Using straight-through instead of crossover.
- Using wrong serial cable (missing DCE/DTE).

Solutions:

- ✓ Use crossover for router-server links.
- ✓ Use straight-through router-switch.
- ✓ Use Serial DCE–DTE between routers.

2. Serial Link Down

Problem: Routers cannot ping across serial interfaces.

Possible Reasons:

- Missing clock rate on DCE side.
- One or both interfaces are shutdown.

Solutions:

- ✓ On DCE: clock rate 64000.
- ✓ Use: no shutdown on both sides.

3. DHCP Failure on PC2

Problem: PC2 gets 169.x.x.x or no IP.

Possible Reasons:

- PC2 left in static mode.
- Wireless router DHCP disabled or wrong LAN interface.

Solutions:

- ✓ PC2 → set to DHCP.
- ✓ Enable DHCP on WRT300N LAN side.

4. Wireless Devices Not Connecting

Problem: PC1/Printer show no Wi-Fi connectivity.

Possible Reasons:

- Wireless NIC not installed.
- SSID/password mismatch.
- Device left powered on during NIC change.

Solutions:

- ✓ Power off device, install WMP300N, power on.
- ✓ Match SSID/security exactly.

5. Cloud–Modem–Router Chain Broken

Problem: Internet-side path shows red or offline.

Possible Reasons:

- Wrong ports used on cloud or modem.
- Incorrect cable type.

Solutions:

- ✓ Use: Cloud Coax7 → Modem0 Port0.
- ✓ Use: Modem0 Port1 → Router Internet port.

6. Router Interfaces Disabled

Problem: Correct IPs but link lights remain red/orange.

Possible Reasons:

- Interface set to OFF in config tab.
- Interface is administratively down.

Solutions:

- ✓ Check 'On' box for each interface.
- ✓ CLI: no shutdown.

7. Missing Static Routes

Problem: LANs cannot reach each other across routers.

Possible Reasons:

- Routers have no routing entries between networks.

Solutions:

- ✓ Router0: ip route 172.16.0.0 255.255.255.0 172.31.0.2
- ✓ Router1: ip route 10.0.0.0 255.255.255.0 172.31.0.1

8. Wrong Cloud0 Port Used

Problem: Router uplink fails even though cable is present.

Possible Reasons:

- Wrong cloud Ethernet port (must be Ethernet6).

Solutions:

- ✓ Use Cloud0 Ethernet6 only.

9. Printer Wi-Fi Setup Fails

Problem: Printer never receives an IP address.

Possible Reasons:

- Wireless NIC missing.
- Wrong SSID/security.
- Printer wireless mode mismatch.

Solutions:

- ✓ Install WMP300N.
- ✓ Match wireless settings exactly.

10. Console Port Not Working

Problem: PC0 cannot access Router0 terminal.

Possible Reasons:

- Wrong PC port used.
- Wrong router port used.

Solutions:

✓ RS-232 from PC0 → Router0 Console.